

NALINA GENKESWARAN

Undergraduate, Computer Science and Engineering,
University of Moratuwa

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ABOUT ME

Second-year Computer Science and Engineering undergraduate at the University of Moratuwa, passionate about backend development and building scalable solutions for real-world problems. A continuous learner who enjoys exploring new technologies, demonstrating leadership, collaborating with teams, and continuously improving technical and professional skills.

PROJECTS

AgriSenseNet (CropWise) – Smart Agriculture Platform – Completed

Docker, Kubernetes, AWS, Terraform, GitHub Actions, ArgoCD, Keycloak, Prometheus, Grafana

Developed a smart agriculture platform for crop monitoring and irrigation optimization using IoT, cloud computing, and AI technologies. As a **Platform, Security & Integration Engineer** (C4 Team), contributed to cloud infrastructure provisioning, Kubernetes orchestration, CI/CD automation, authentication and authorization, security, and observability by implementing a scalable deployment architecture.

An AI-Powered Technical Debt Analytics Dashboard for Agile Small-Scale Software Teams- In Progress

Spring Boot, React.js, Python, Tree-sitter, Lizard, PostgreSQL, RabbitMQ, Machine Learning

Developing a full-stack SaaS platform for automated software repository analysis using GitHub integration, Tree-sitter, Lizard, and Machine Learning. The system identifies Self-Admitted Technical Debt (SATD), predicts bug-prone files, calculates repository health and technical debt scores, and generates prioritized refactoring recommendations through an analytics dashboard.

Offline-First Emergency Communication & Disaster Prioritization System – Completed

Spring Boot, PostgreSQL, React.js, Google Gemini API, SMSMobile API, REST APIs

- Developed an offline-first emergency communication and disaster management system that collects emergency reports through mobile applications and SMS, ensuring reliable communication during disasters and network disruptions.
- Contributed as a **Backend Developer** in a 4-member team, developing REST APIs, database operations, SMS integration, AI-based severity analysis, disaster prioritization, and volunteer matching for efficient emergency response.

Rescue Team Management System (Individual Project) – Completed

Spring Boot, Java, React.js, MySQL, Spring Data JPA, REST APIs

Developed a full-stack disaster and rescue team management system independently, implementing RESTful APIs, database design, and role-based dashboards to support emergency reporting, disaster verification, rescue team assignment, status tracking, and incident resolution workflows.

Microbanking & Interest Management System (MIMS) - Completed

Next.js, PostgreSQL, JWT, REST APIs, Tailwind CSS

- Developed a full-stack banking application supporting customer management, transactions, fixed deposits, automated interest distribution, and financial reporting.
- Collaborated in a 5-member team to implement responsive user interfaces, secure authentication, RESTful APIs, and PostgreSQL-based data management for banking operations.

RPAL Interpreter - Completed

C++, Compiler Design, AST, Recursive Descent Parsing

- Developed an interpreter for the RPAL programming language, implementing lexical analysis, parsing, AST generation, and program execution.
- Collaborated in a 2-member team to build a CSE machine for evaluating standardized RPAL programs.

EDUCATION

Bsc(Hons) in Computer Science and Engineering

University of Moratuwa.

CGPA - 3.45



Feb 2024 - Ongoing

GCE Advanced Level - 2022

Vembadi Girl's High School, Jaffna.

Z.Score - 2.2574

School Rank - 1st

District rank - 23

Island rank - 525



2020 - 2022

EXPERIENCE

Frontend Developer | Mora Exams Community



2023, 2025

- Developed frontend features for the Mora Exams Website and Admin Dashboard supporting island-wide A/L examinations.
- Played an active role in the planning, coordination, and successful execution of large-scale examinations, working with cross-functional teams to manage operations and participant engagement across Sri Lanka.

SKILLS

Technical

- **Programming Languages:** Java, Python, C+ +, C, JavaScript/TypeScript, SQL
- **Frameworks & Libraries:** Spring Boot, React, Node.js, Express
- **Databases:** MySQL, PostgreSQL, , MongoDB
- **Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, Nginx
- **Tools & Technologies:** Git, GitHub, Kafka, Postman, Linux

Languages

- **Tamil** - Native
- **English** - Fluent

ACHIEVEMENTS

- **IAEA Nuclear Science App Challenge 2026** - Sri Lanka Representative (University Category Winner)
- **CryptX Hackathon 2026** - Finalist
- **UOJ Coders Hackathon 2026** - Finalist

CERTIFICATIONS

- Java Programming by simplilearn (December 2025)
- Intro to Machine learning by kaggle (January 2026)
- Data cleaning by kaggle (February 2026)
- Pandas by kaggle (February 2026)
- SQL by HackerRank ((December 2025)

REFEREES

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